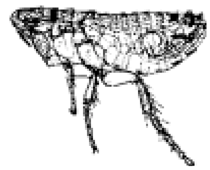
flea
NEWS 86



September 2020

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FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Assistant Editor J. R. Kucera, M.S.

Contents

Editorial

More on Keystone Parasites

Dear *Flea News* Reader,

In the last Flea News, the topic was "keystone fleas". Why study keystone parasites?

Crozier and Schulte-Hostedde (2014) commented, "One consideration that has not received sufficient attention is that a conservation mandate to protect biodiversity might imply the need to protect pathogens from extinction in part because of their ecological role as consumers. Insofar as wildlife disease eradication programs are aimed at minimizing or extirpating the organism directly responsible for mortality and morbidity in wildlife, livestock, and human populations, they might undermine biodiversity conservation". Acosta and Fernández (2015) wrote, "Parasite conservation may not be a popular topic, but preserving and studying parasite diversity and interactions represent many benefits... for actively modeling community structure and keeping diversity, for being good indicators of ecosystem health..., for their use in biomedicine, and because their DNA may provide a biological record of evolutionary dynamics...".

Some parasites are extinct or endangered because their host went extinct, or because the host's conservation eliminated the host's ectoparasites (Rózsa and Vas 2015). They note, "… several reasons why conservationists should care about threatened parasites. They not only constitute a large proportion of global biodiversity but also exert selective pressures to increase host diversity... and therefore harbouring a unique parasitic fauna can increase the conservation value of the host... Furthermore, parasites carry phylogenetic and population genetic information about the evolutionary past of their hosts... Conservationists should consider preserving host specific lice as part of their efforts to save birds or mammals ex situ" (Rózsa and Vas 2015).

Keystone parasites and their pathogens, including viruses ("virosphere"), construct "pathosystems". The term "pathosystem" originated with plant-pathogen systems (Robinson 1987), but applies equally to animal-pathogen systems. Pathosystems are defined as an ecosystem's subsystems of parasites and disease.

Keystone parasite effects have been examined in diverse pathosystems, from microsporidians infecting caddisflies, to nematodes and viruses found in deer mice (Tompkins et al. 2011). Keystone parasites are particularly influential in determining the course of biological invasions, when invading species may escape their usual predators and parasites ("Enemy Release Hypothesis") (Telfer et al. 2005, Dunn 2009, Mlynarek 2015, Gozzi et al. 2020).

The predicted negative effect of parasites on hosts is formalized as the "Ectoparasite Release Hypothesis" (EPRH). Simply put, EPRH postulates that a host benefits when their ectoparasites are removed (Kluever et al. 2019). Examples abound, such as ectoparasitic mites, *Echinonyssus blanchardi*, which increase their marmot host's weight loss during hibernation (Arnold and Anja 1993).

However, positive effects of parasites on their hosts and other parasites exist, too (Veitch et al. 2020). This is especially true when we consider indirect effects. Mlynarek (2015) found parasitic water-mites and gregarines attacked invasive damselflies which acted potentially as a "sink" for the parasites, possibly benefiting an already established damselfly.

In both terrestrial and aquatic ecosystems, "parasites - usually regarded in terms of their detrimental effects on the individuals they infest - can also have positive impacts... parasites and pathogens act as ecological engineers... [and] challenge the conventional wisdom that parasites have only negative or inconsequential impacts on ecological communities" (Hatcher et al. 2012, 2014). The impact of parasites on ecosystem processes has led to calls to include parasites in food web and ecosystem studies (Pojmariska 2002, Hudson et al. 2006, Byers 2009).

Keystone parasites are likely essential regulatory elements in their communities, but these processes are largely unstudied (Watson 2013, Sponchiado et al. 2017). These unknowns call into question the unintended consequences of global programs designed to eliminate disease, such as malaria (Small 2019, Anon. 2020).

Well wishes in regard to the coronavirus pandemic.

Yours in fleas,
R.L. Bossard
Editor, *Flea News*

Literature

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Telfer, S., Brown, K.J., Sekules, R., Begon, I., Hayden, T. & Birtel, R. (2005) Disruption of a host-parasite system following the introduction of an exotic host species. Parasitology, 130, 661–665.

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Watson, Maggie J. 2013. What drives population-level effects of parasites? Meta-analysis meets life-history. International Journal for Parasitology: Parasites and Wildlife 2 (2013) 190–196.

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Announcement

Congratulations to Dr. Michael Dryden upon his retirement to emeritus faculty at Kansas State University. His email address will remain active.

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Featured Research

New Flea Species 2020

Hastriter, M. W. (2020). Fleas (Siphonaptera: Pygiopsyllomorpha) of Papua New Guinea and Papua Province (Indonesia). Part V. *Astivalius, Idiochaetis, Muesebeckella*, and *Obtusifrontia* (Stivaliidae: Stivaliinae), with Description of Six New Species. Annals of Carnegie Museum, 86(2), 169-196.

The genera *Astivalius* Smit, 1953, and *Obtusifrontia*, Holland, 1969 endemic to Papua Province, Indonesia, and *Astivalius, Idiochaetis* Jordan, 1937, *Muesebeckella* Traub, 1969, and *Obtusifrontia,* endemic to Papua New Guinea, are reviewed as a continuation of the study of fleas in the Robert Traub flea collection deposited in the Carnegie Museum of Natural History, Pittsburgh, PA. This paper (Part V) is an extension of previous studies by Hastriter (2012), Hastriter and Easton (2013, Part I, *Striopsylla*), Hastriter (2014, Part II, *Nestivalius, Orthopsylloides*, and *Parastivalius*), Hastriter (2015, Part III, *Traubia*), and Hastriter (2016, Part IV, *Rectidigitus*). Prior to the current study, *Astivalius* and *Idiochaetis* were each comprised of one species (*A. microphthalmus* Smit, 1953, and *I. illustris* Jordan, 1937), *Muesebeckella* of two species (*Mu. mannae* Traub, 1969, and *Mu. nadi* Traub, 1969), and *Obtusifrontia* of three species (*O. falcata* Mardon, 1978b, *O. simplex* Holland, 1969, and *O. simula* Mardon, 1978b). The female of *A. microphthalmus* Smit, 1953, is described for the first time and the previously known distribution of this species is expanded from Papua Province, Indonesia to the western fringes of Papua New Guinea (Sandaun Province). The female of *O. simplex* is also described for the first time and its geographical distribution is expanded to two additional provinces (Southern Highlands and Western Highlands provinces, Papua New Guinea). An additional three new species of *Astivalius*, one new species each of *Idiochaetis, Muesebeckella*, and *Obtusifrontia* are described herein (*A. archboldi, n. sp., A. mirzai, n. sp., A. toxopeusi, n. sp., I. rogersi, n. sp., Mu. niobiensis, n. sp., and O. comohamulus, n. sp.*). With the description of six new species, the total number of species in the superfamily Pygiopsylloidea in Papua Province, Indonesia, Papua New Guinea (including Bismarck Archipelago) and the Solomon Islands is 111. An additional eight species belonging to three other flea families (Ischnopsyllidae (3), Pulicidae (3), and Leptopsyllidae (2) bring the total number of flea taxa to 119 species (including subspecies). Keys to the species of *Astivalius, Idiochaetis, Muesebeckella*, and *Obtusifrontia* are provided.

[*A. archboldi, n. sp., A. mirzai, n. sp., A. toxopeusi, n. sp., I. rogersi, n. sp., Mu. niobiensis, n. sp., and O. comohamulus, n. sp.* Hastriter 2020]

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Keskin, A. (2020). A new flea species of the genus *Palaeopsylla* (Insecta: Siphonaptera: Ctenophthalmidae) from Turkey. Journal of Medical Entomology, 57(1), 88-91.

A new flea (Insecta: Siphonaptera) species belonging to the *Palaeopsylla minor*-group, *Palaeopsylla (Palaeopsylla) beaucournui* n. sp. is described and illustrated. Specimens (14 males and seven females) of this new species were collected from *Talpa levantis* Thomas (Mammalia: Talpidae) in the deciduous mixed forest in Bursa province of Marmara Region, Turkey.

[*Palaeopsylla (Palaeopsylla) beaucournui* Keskin 2020]

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López-Berrizbeitia, M. F., Pérez, M. J., & Barquez, R. M. (2020). Description of two new species of *Ectinorus (Ectinorus)* (Siphonaptera, Rhopalopsyllidae) from Argentina, including a morphometric approach. Acta Tropica, 211, 105612.

Two new species of fleas of genus *Ectinorus* (Siphonaptera: Rhopalopsyllidae) are described from sigmodontine rodents (Rodentia: Cricetidae) collected during a survey of small mammals in northwestern Argentina. The new species belong to the subgenus *Ectinorus* and can be distinguished from all other species of the subgenus by the characteristics of the modified abdominal segments and by the genitalia. Moreover, the male of *Ectinorus (Ectinorus) disjugis* is described for the first time and the finding of this flea parasitizing the rodent *Akodon spegazzinii,* constitutes a new flea–host association. A phylogenetic analysis of the genus *Ectinorus*, using traditional morphological characters and morphogeometric data, is presented to support the erection of the new species of *Ectinorus*. An identification key for all species of *Ectinorus* is also provided. Our study increases to 38 the total number of species of the subgenus *Ectinorus*, and to 20 the number for Argentina. Phylogenetic analyses reveal that *Ectinorus* is monophyletic but the subgenera are not. This study offers a new interpretation of morphological diversity within the genus as well as an evaluation of hypotheses about their relationships.

Taxonomy

Hastriter, M. W., & Eckerlin, R. P. (2020). Report of a Small Collection of *Kohlsia mojica* Tipton and Méndez, 1966 (Siphonaptera: Ceratophyllidae) from Costa Rica with a Discussion of Several Enigmatic Species of *Kohlsia* Traub, 1950. Proceedings of the Entomological Society of Washington, 122(3), 632-649.

A collection of fleas was acquired independently from Costa Rica by two different field teams. Among these collections were a sizable series of *Kohlsia mojica* Tipton and Méndez which were previously reported only from the northern province of Bocas del Toro, Panama. These new records were present in two provinces of Costa Rica (Alajuela and Puntarenas). Additional specimens discovered in the Robert Traub flea collection in the Carnegie Museum of Natural History, heretofore unreported, included specimens from San Jose and Cartago provinces, Costa Rica. The primary host for *K. mojica* is the Mexican deer mouse, *Peromyscus mexicanus* Saussure. Collectively, five species of *Kohlsia* have been reported within the geographical boundaries of Costa Rica and Panama [*K. mojica, K. azuerensis* Tipton and Méndez, *K. keenani* Tipton and Méndez, *K. graphis graphis* (Rothschild), and *K. traubi* Tipton and Méndez]. A key is provided for these five species. Specimens of *K. g. graphis* are reported herein for the first time from Guatemala and Mexico, and the variegated squirrel *Sciurus variegatoides* Ogilby represents a new host record for *K. g. graphis*. Table 1 is included to illustrate the distribution of all known *Kohlsia* species. In addition, *Kohlsia campaniger* Jordan (an extralimital species) described from a single female specimen from Ecuador is discussed. Several morphological characters are provided to distinguish *K. campaniger* and *Kohlsia felteni* Smit. *Jellisonia ortizi* Vargas (currently valid in *Kohlsia*) is hypothesized to be a new synonym of *Kohlsia fournieri* Vargas. *Kohlsia keenani* is reported for the first time from *Handleyomys alfaroi* (J. A. Allen), *Handleyomys chapmani* (Thomas) and *Nephelomys albigularis* (Tomes).

Kwak, M. L., & Hastriter, M. W. (2020). The Australian giant fleas *Macropsylla* Rothschild, 1905 (Siphonaptera: Macropsyllidae: Macropsyllinae), their identification, evolution, ecology, and conservation biology. Systematic Parasitology, 97(1), 107-118.

Macropsylla Rothschild, 1905 is an endemic Australian flea genus represented by two species: *M. hercules* Rothschild, 1905 and *M. novaehollandiae* Hastriter, 2002. However, their identification is challenging. To address this difficulty, an extensive differential diagnosis for the two species is provided along with a key to distinguish *Macropsylla* from other Australian flea genera. The first record of *M. hercules* from the domestic cat (*Felis catus* (L.)) is also presented. The taxonomy, distribution, host relationships, evolution, and ecology of the *Macropsylla* species are discussed, along with the conservation biology of the threatened New Holland flea *M. novaehollandiae*.

LÓPEZ-BERRIZBEITIA, M. F., Sanchez, J., BARQUEZ, R., & Diaz, M. (2020). Taxonomic revision of the flea genus *Agastopsylla* Jordan & Rothschild 1923 (Siphonaptera: Ctenophthalmidae). Anais da Academia Brasileira de Ciências, 92(1).

Thoroughgood, J. T., Armstrong, J. S., White, B., Anstead, C. A., Galloway, T. D., Lindsay, L. R., ... & Chilton, N. B. (2020). Molecular differentiation of four species of *Oropsylla* (Siphonaptera: Ceratophyllidae) using PCR-based single strand conformation polymorphism analyses and DNA sequencing. Journal of Medical Entomology.

Zurita, A., García-Sánchez, Á. M., & Cutillas, C. (2020). *Ctenophthalmus baeticus boisseauorum* (Beaucournu, 1968) and *Ctenophthalmus apertus allani* (Smit, 1955)(Siphonaptera: Ctenophthalmidae) as synonymous taxa: morphometric, phylogenetic, and molecular characterization. Bulletin of

Entomological Research, 1-14.

The family Ctenophthalmidae (Order Siphonaptera) has been considered as a ‘catchall’ for a wide range of divergent taxa showing a paraphyletic origin. In turn, *Ctenophthalmus* sp. (Ctenophthalmidae) includes 300 valid described taxa. Within this genus, males are easily distinguishable basing on the size, shape, and chaetotaxy of their genitalia; however, females show slight morphological differences with each other. The main objective of this work was to carry out a comparative morphometric, phylogenetic, and molecular study of two different subspecies: *Ctenophthalmus baeticus boisseauorum* and *Ctenophthalmus apertus allani* in order to clarify and discuss its taxonomic status. From a morphological and biometrical point of view, we found clear differences between modified abdominal segments of males of both subspecies and slight differences in the margin of sternum VII of all female specimens which did not correspond with molecular and phylogenetic results based on four different molecular markers (Internal Transcribed Spacer 1 and 2 of ribosomal DNA, and the partial cytochrome c oxidase subunit 1 and cytochrome b of mitochondrial DNA). Thus, we observed a phenotypic plasticity between both subspecies, which did not correspond with a real genotypic variability nor different environmental or ecological conditions. Basing on these results, we could consider that there are no solid arguments to consider these two ‘morphosubspecies’ as two different taxa. We propose that *C. b. boisseauorum* should be considered as a junior synonym of *C. a. allani*.

Hosts and Distribution

Acosta, R., Guzman-Cornejo, C., CISNEROS, F. A. Q., QUIÑONEZ, A. A. T., & Fernandez, J. A. (2020). New records of ectoparasites for Mexico and their prevalence in the montane shrew *Sorex monticolus* (Eulipotyphla: Soricidae) at Cerro del Mohinora, Sierra Madre Occidental of Chihuahua, Mexico. Zootaxa, 4809(2), 393-396.

Baláž, I., & Zigová, M. (2020). Flea Communities on Small Mammals in Lowland Environment. Ekológia (Bratislava), 39(3), 260-269.

Bergman, C. M., Galloway, T. D., & Sinkins, P. (2020). Collections of fleas (Siphonaptera) from Pacific marten, *Martes caurina* (Carnivora: Mustelidae), reveal unique host–parasite relationships in the Haida Gwaii archipelago. Journal of the Entomological Society of British Columbia, 116, 17-24.

Fleas and their host–parasite relationships are understudied in many parts of Canada, yet such relationships may contribute to our knowledge of ecosystems in ways we have yet to understand. A collection of 57 fleas from Pacific marten (*Martes caurina* (Merriam)) in Haida Gwaii, off the coast of British Columbia, Canada, led to the collection of five taxa of fleas: the European rat flea, *Nosopsyllus fasciatus* (Bosc), a squirrel flea, *Ceratophyllus (Amonopsyllus) ciliatus protinus* (Jordan), a mustelid flea, *Chaetopsylla floridensis* (I. Fox), *Hystrichopsylla (Hystroceras) dippiei*, likely ssp. *Spinata* Holland, a parasite of mustelids and mephitids, and a generalist bird flea, *Dasypsyllus gallinulae*

perpinnatus (Baker). All five species are first records for Haida Gwaii, and *C. floridensis* is recorded from Canada for the first time. Two new host–parasite relationships support a previous dietary study of marten in Haida Gwaii. This provides further evidence that fleas infesting predators may indicate prey composition within their home ranges.

Horb, K.O. (2020). BREED SUSCEPTIBILITY OF DOGS TO ECTOPARASITES IN THE GENUS *CTENOCEPHALIDES* (SIPHONAPTERA, PULICIDAE). Bulletin of Poltava State Agrarian Academy, (2), 164-169. [Горб, К. О. (2020). ПОРОДНА СПРИЙНЯТЛИВІСТЬ ДОМАШНІХ СОБАК ДО ЕКТОПАРАЗИТІВ РОДУ *CTENOCEPHALIDES* (SIPHONAPTERA, PULICIDAE). Вісник Полтавської державної аграрної академії, (2), 164-169.]

Hernández-Urbina, C. F., Vital-García, C., Ávila, A. M. E., Colima, A. G., Sánchez-Olivas, M. P., & Clemente-Sánchez, F. (2020). First report of Siphonaptera parasites in *Canis latrans* in the Flora and Fauna Protection Area, Médanos de Samalayuca Chihuahua, Mexico. Veterinary Parasitology: Regional Studies and Reports, 20, 100379.

KESKIN, A., & BEAUCOURNU, J. C. (2020). Updated list of fleas (Insecta: Siphonaptera) of the Korean Peninsula, with the presence of *Rhadinopsylla alphabetica* Jameson & Sakaguti in the Democratic People's Republic of Korea. Zootaxa, 4838(2), 248-256.

Keskin, A., Selçuk, A. Y., Kefelioğlu, H., & Beaucournu, J. C. (2020). Fleas (Insecta: Siphonaptera) collected from some small mammals (Mammalia: Rodentia, Eulipotyphla) in Turkey, with new records and new host associations. Acta Tropica, 105522.

Fleas (Insecta: Siphonaptera) are among the most common ectoparasites of small mammals. We investigated fleas infesting on the small mammals (Mammalia: Rodentia, Eulipotyphla) from 15 different localities in Turkey. A total of 276 flea (133 males and 143 females) specimens belonging to 32 different flea taxa were collected from 90 (42 males, 16 females and 32 undetermined) small mammals belonging to 20 different species. With many new flea-host associations, the first occurrences of *Ctenophthalmus agyrtes peusianus* Rosicky, *Ctenophthalmus golovi golovi* Ioff & Tiflov, *Ctenophthalmus secundus vicarius* Jordan & Rothschild, *Doratopsylla dasycnema cuspis* Rothschild, *Leptopsylla algira popovi* (Wagner & Argyropulo) and *Rhadinopsylla pentacantha* (Rothschild) fleas were reported in Turkey. In addition, three flea species, *Ctenophthalmus coniunctus* Peus, *Ctenophthalmus contiger* Peus, and *Palaeopsylla incisa* Peus, are reported for only the second time since their original descriptions.

Keskin, A., Dik, B., Karatepe, M., & Karatepe, B. (2020). The presence of little-known flea *Caenopsylla laptevi laptevi* (Insecta: Siphonaptera) in Turkey with a re-description outside the type locality. Transactions of the American Entomological Society, 146(1), 258-264.

Litvinov, M. N., Litvinova, E. A., Erofeeva, M. N., & Naidenko, S. V. (2020). Parasitic Community of Fleas (Siphonaptera) of Small and Medium Predators (Mammalia, Carnivora) of Southwestern Primorskii Krai. CONTEMPORARY PROBLEMS OF ECOLOGY, 13(2), 180-183.

Shu, C., Jiang, M., Yang, M., Xu, J., Zhao, S., Yin, X., ... & Wang, Y. (2020). Flea surveillance on wild mammals in northern region of Xinjiang, northwestern China. International Journal for Parasitology: Parasites and Wildlife, 11, 12-16.

Wang, H. C., Lee, P. L., & Kuo, C. C. (2020). Fleas of Shrews and Rodents in Rural Lowland Taiwan. Journal of Medical Entomology, 57(2), 595-600.

Physiology

Almeida, G. P. S. D., Campos, D. R., Avelar, B. R. D., Silva, T. X. D. A. D., Lambert, M. M., Alves, M. S. R., & Correia, T. R. (2020). Development of *Ctenocephalides felis felis* (Siphonaptera: Pulicidae) in different substrates for maintenance under laboratory conditions. Revista Brasileira de Parasitologia Veterinária, 29(2).

The aim of this study was to evaluate the efficiency of different substrates for larval development of *Ctenocephalides felis felis* during its biological cycle. Eight hundred eggs of *C. felis* felis from a flea maintenance colony were used. Different diets were formulated, in which the main substrates were meat flour, powdered milk, sugar, lyophilized bovine blood, tick metabolites and lyophilized egg. The flea eggs were placed in test tubes (10 per tube) and approximately 2 g of the diet to be tested was added to each tube. There were 10 replicates for each substrate. After 28 days, each tube was evaluated individually for the presence of pupae and emerged adults. The following percentages of the larvae completed the cycle to the adult stage: 67% in diets containing tick metabolites; 55%, meat flour; 39%, dehydrated bovine blood; 14%, powdered milk; and less than 1% in diets containing sugar, lyophilized bovine blood, lyophilized egg or wheat bran. It was concluded that among the diets tested, the one constituted by tick metabolites as the substrate was shown to be the most satisfactory for maintaining a laboratory colony of *C. felis felis*, followed by the one containing meat flour.

Chao Wan, Zhixiu Hao. 2020. Natural arrangement of micro-strips reduces shear strain in the locust cuticle during power amplification. J Biomech 109842.

Horb, K.O. 2020. Biochemical parameters of blood serum of dogs for *Ctenocephalides*. Sci Messenger of LNU of Vet Med and Biotech. Series: Vet Sci 22(97): 3-6.

Evolution

Azrizal-Wahid, N., Sofian-Azirun, M., & Low, V. L. (2020). New insights into the haplotype diversity of the cosmopolitan cat flea *Ctenocephalides felis* (Siphonaptera: Pulicidae). Veterinary Parasitology,

109102.

Miao Y, Hua B-Z (2020) The highly rearranged karyotype of the hangingfly *Bittacus sinicus* (Mecoptera, Bittacidae): the lowest chromosome number in the order. CompCytogen 14(3): 353–367 (2020) doi: 10.3897/CompCytogen.v14i3.53533 http://compcytogen.pensoft.net

Cytogenetic features of the hangingfly *Bittacus sinicus* Issiki, 1931 were investigated for the first time using C-banding and DAPI (4',6-diamidino-2-phenylindole) staining. The karyotype analyses show that the male *B. sinicus* possesses the lowest chromosome number (2n = 15) ever observed in Mecoptera, and an almost symmetric karyotype with MCA (Mean Centromeric Asymmetry) of 12.55 and CVCL (Coefficient of Variation of Chromosome Length) of 19.78. The chromosomes are either metacentric or submetacentric with their sizes decreasing gradually. Both the C-banding and DAPI+ patterns detect intermediate heterochromatin on the pachytene bivalents of *B. sinicus*, definitely different from the heterochromatic segment at one bivalent terminal of other bittacids studied previously. The male meiosis of *B. sinicus* is chiasmate with two chiasmata in metacentric bivalents and one in the submetacentric bivalent. The sex determination mechanism is X0(♂), which is likely plesiomorphic in Bittacidae. Two alternative scenarios of karyotype origin and evolution in *Bittacus* Latreille, 1805 are discussed.

Keywords C-banding technique, chromosome rearrangement, cytogenetics, DAPI, evolution, Holometabola, meiosis

Nascimento, C. S. I., Moura, J. F., Robbi, B., & Fernandes, M. A. (2020). Lesions in osteoderms of pampatheres (Mammalia, Xenarthra, Cingulata) possibly caused by fleas. Acta Tropica, 211, 105614.

In this study, the first records of lesions in osteoderms of *Holmesina,* a group of fossil cingulates related to armadillos, possibly caused by the action of penetrating fleas (Siphonaptera) are described. Three individuals of *Holmesina cryptae* (Pampatheriidae) were collected from Quaternary sediments in Lapinha Cave (Iramaia, Bahia state, Brazil). Their osteoderms were analyzed by stereomicroscope and scanning electron microscopy and alterations on their surfaces were recognized. We found 63 marks distributed in 23 of 1300 analyzed osteoderms (approximately 1.8% of the total of osteoderms), characterized by vertical cavities with well-delimited circular borders similar to those lesions made by Tungidae fleas in extant mammals. These records indicate that there was an interaction between penetrating fleas and pampatheres during the Quaternary in Brazilian Intertropical Region, and contribute to the understanding of the evolution of these ectoparasites and the relationship with their hosts.

Zhao, X., Wang, B., Bashkuev, A. S., Aria, C., Zhang, Q., Zhang, H., ... & Engel, M. S. (2020). Mouthpart homologies and life habits of Mesozoic long-proboscid scorpionflies. Science Advances, 6(10), eaay1259.

Zhang, Y., Shih, C., Rasnitsyn, A. P., Ren, D., & Gao, T. (2020). A new Early Cretaceous flea from China. Acta Palaeontologica Polonica, 65(1), 99-107.

Fleas are highly specialized holometabolic insects. So far, only 16 species of fossil fleas in five families have been reported due to the rare fossil records. At present, the earliest flea fossils are reported from the Middle Jurassic Jiulongshan Formation of Northeastern China. The descriptions of these earliest species pushed back the origin of Siphonaptera by at least 40 million years. It is generally accepted that saurophthirids are the “transitional” taxa from stem Jurassic fleas to living crown groups. Herein, we described a new “transitional” flea species, *Saurophthirus laevigatus* Zhang, Shih, Rasnitsyn, and *Gao sp. nov.*, from the Lower Cretaceous Yixian Formation of Northeastern China, assigned to Saurophthiridae. The new species provides new evidence to support saurophthirids as a “transitional” group. Sexual dimorphism suggests significant differences in biology of opposite sexes in *Saurophthirus*. Analysis of described Mesozoic species demonstrates the body size reduction from the Middle Jurassic to the Early Cretaceous. Smaller body size was likely advantageous in reducing the probability of being detected and removed by the host and in minimizing flea’s demand for blood intake and energy input, indicating the adaptation of the ectoparasitic lifestyle of fleas in their early stage of evolution.

Ecology

Gozzi A.C., Marcela Lareschi, G. Navone, M.L. Guichón. 2020. The Enemy Release Hypothesis and *Callosciurus erythraeus* in Argentina: combining community and biogeographical parasitological studies. Biological Invasions. DOI 10.1007/s10530-020-02339-w

Krasnov, B. M. Stanko, M. Lareschi, I. S. Khokhlova. 2020. Species co-occurrences in ectoparasite infracommunities: accounting for confounding factors associated with space, time, and host community composition. Ecological Entomology 45: 1158-1171. DOI:10.1111/een.12900

Maestri, Renan, Maico S. Fiedler, Georgy I. Shenbrot, Elena N. Surkova, Sergei G. Medvedev, Irina S. Khokhlova, Boris R. Krasnov. 2020. Harrison’s rule scales up to entire parasite assemblages but is determined by environmental factors. Journal of Animal Ecology First published: 16 September https://doi.org/10.1111/1365-2656.13344

Harrison’s rule states that parasite body size and the body size of their hosts tend to be positively correlated. After it was proposed a century ago, a number of studies have investigated this trend, but the support level has varied greatly between parasite/host associations. Moreover, while the rule has been tested at the individual species level, we still lack knowledge on whether Harrison’s rule holds at the scale of parasite and host communities. Here, we mapped flea (parasites) and rodent (hosts) body sizes across Mongolia and asked whether Harrison’s rule holds for parasite/host assemblages (i.e., whether a parasite’s average body size in a locality is positively correlated with its host’s average body size). In addition, we attempted to disentangle complex relationships between flea size, host size, and environmental factors by testing alternative hypotheses for the determinants of fleas’ body size variation. We gathered occurrence data for fleas and rodents from 2370 sites across Mongolia, constructed incidence matrices for both taxa, and calculated the average body sizes of fleas and their hosts over half-degree cells. Then, we applied a path analysis, accounting for spatial autocorrelation, trying to disentangle the drivers of the correlation between parasite and host body sizes. We found a strong positive correlation between average flea and host size across assemblages. Surprisingly though, we found that environmental factors simultaneously affected the body sizes of both fleas and hosts in the same direction, leading to a most likely deceptive correlation between parasite and host size across assemblages. We suggest that environmental factors may, to a great extent, reflect the environmental conditions inside the hosts' burrows where fleas develop and attain their adult body size, thus influencing their larval growth. Similarly, rodent body size is strongly influenced by air temperature, in the direction predicted by Bergmann's rule. If our findings are valid in other host-parasite associations, this may explain the dissenting results of both support and lack thereof for Harrison's rule.

Veitch, Jasmine S.M., Jeff Bowman, Albrecht I. Schulte-Hostedde. 2020. Parasite species co-occurrence patterns on *Peromyscus:* Joint species distribution modelling. IJP: Parasites and Wildlife 12 (2020) 199–206.

Hosts are often infested by multiple parasite species, but it is often unclear whether patterns of parasite cooccurrence are driven by parasite habitat requirements or parasite species interactions. Using data on infestation patterns of ectoparasitic arthropods (fleas, trombiculid mites, cuterebrid botflies) from deer mice (*Peromyscus maniculatus*), we analyzed species associations using joint species distribution modelling. We also experimentally removed a flea (*Orchopeas leucopus*) from a subset of deer mice to examine the effect on other common ectoparasite species. We found that the mite (*Neotrombicula microti*) and botfly (*Cuterebra* sp.) had a negative relationship that is likely a true biotic species interaction. The flea had a negative association with the mite and a positive association with the botfly species, both of which appeared to be influenced by host traits or parasite life-history traits. Furthermore, experimental removal of the flea did not have a significant effect on ectoparasite prevalence of another species. Overall, these findings suggest that complex parasite species associations can be present among multiple parasite taxa, and that aggregation is not always the rule for ectoparasite communities of small mammals.

Veitch, Jasmine S.M. 2020. Ectoparasitism of rodent hosts in Algonquin Provincial Park, Ontario, Canada: Infestation patterns, host glucocorticoids, and species co-occurrence. Master of Science (MSc) in Biology, thesis. Laurentian University, Sudbury, Ontario, Canada.

Examining multiple parasite taxa across host species presents an opportunity to assess the biology of host-parasite systems. This study investigated: 1) factors associated with ectoparasite prevalence on deer mice (*Peromyscus maniculatus*), southern red-backed voles (*Myodes gapperi*) and woodland jumping mice (*Napaeozapus insignus*); 2) relationship between ectoparasites and glucocorticoid levels of deer mice, and; 3) whether ectoparasites of deer mice and North American red squirrels (*Tamiasciurus hudsonicus*) form structured assemblages. I examined data from fleas, mites, and botflies on these hosts in Algonquin Provincial Park, Ontario, Canada. Ectoparasite prevalence varied with host traits and date. Ectoparasites had no relationship with deer mouse glucocorticoid production. Lastly, ectoparasites of deer mice, but not red squirrels, had exhibited non-random co-occurrence patterns. Parasites play an important role in population regulation and thus, these findings provide a better understanding on the effect of ectoparasites on their hosts, on each other, and consequently on their ecosystem.

Keywords: ectoparasites; rodents; fecal glucocorticoid metabolites; corticosterone; species cooccurrence.

Warburton, E.M., Untapped potential: The utility of drylands for testing ecoevolutionary relationships between hosts and parasites, International Journal for Parasitology: Parasites and Wildlife (2020), doi: https://doi.org/10.1016/j.ijppaw.2020.04.003.

Pathology and Control

El Hamzaoui, Basma, Antonio Zurita, Cristina Cutillas, Philippe Parola. 2020. Fleas and flea-borne diseases of North Africa. Acta Tropica 211 (2020) 105627. https://doi.org/10.1016/j.actatropica.2020.105627

North Africa has an interesting and rich wildlife including hematophagous arthropods, and specifically fleas, which constitute a large part of the North African fauna, and are recognised vectors of several zoonotic bacteria. Flea-borne organisms are widely distributed throughout the world in endemic disease foci, where components of the enzootic cycle are present. Furthermore, flea-borne diseases could re-emerge in epidemic form because of changes in the vector–host ecology due to environmental and human behaviour modifications. We need to know the real incidences of flea-borne diseases in the world due to this incidence could be much greater than are generally recognized by physicians and health authorities. As a result, diagnosis and treatment are often delayed by health care professionals who are unaware of the presence of these infections and thus do not take them into consideration when attempting to determine the cause of a patient's illness. In this context, this bibliographic review aims to summarise the main species of fleas present in North Africa, their geographical distribution, fleaborne diseases, and their possible re-emergence.

Erkunt Alak, S., Köseoğlu, A.E., Kandemir, Ç. et al. High frequency of knockdown resistance mutations in the para gene of cat flea (Ctenocephalides felis) samples collected from goats. Parasitol Res (2020). https://doi.org/10.1007/s00436-020-06714-3

Fleas are ectoparasites of mammals and birds. In livestock such as sheep and goat, flea bites cause many clinical signs. Several types of insecticides including pyrethroids are used to struggle against fleas. The widespread use of these insecticides causes an increase in the number of resistant individuals in flea populations. T929V and L1014F mutations corresponding to pyrethroid resistance have been found in the para gene of cat fleas. We aimed to investigate T929V and L1014F mutations in flea samples (n:162) collected from goats in seven different farms where cypermethrin, a synthetic pyrethroid, had been used intensively. To achieve this aim, collected flea samples were morphologically identified under a stereo microscope and DNA isolation was conducted by HotSHOT method. Later, a bi-PASA targeting the para gene was applied to identify both mutations in corresponding samples. According to the results obtained, all fleas were *Ctenocephalides felis*. Frequencies of T929V and L1014F mutations in fleas were 92.6% (150/162) and 95.7% (155/162), respectively. In conclusion, the frequency of mutations related to pyrethroid resistance was very high in the fleas collected from all the farms and it was thought that the high frequency of these mutations can be attributed to intensive use of pyrethroids.

Greigert, V., Brunet, J., Ouarti, B., Laroche, M., Pfaff, A. W., Henon, N., ... & Abou-Bacar, A. (2020). The Trick of the Hedgehog: Case Report and Short Review About *Archaeopsylla erinacei* (Siphonaptera: Pulicidae) in Human Health. Journal of Medical Entomology, 57(1), 318-323.

Harimalala, M., Rahelinirina, S., & Girod, R. (2020). Presence of the Oriental Rat Flea (Siphonaptera: Pulicidae) Infesting an Endemic Mammal and Confirmed Plague Circulation in a Forest Area of Madagascar. Journal of Medical Entomology.

Kotti, B. K., & Zhilzova, M. V. (2020). The Significance of Fleas (Siphonaptera) in the Natural Plague

Foci. Entomological Review, 100(2), 191-199.

Lemon, A., Cherzan, N., & Vadyvaloo, V. (2020). Influence of Temperature on Development of *Yersinia pestis* Foregut Blockage in *Xenopsylla cheopis* (Siphonaptera: Pulicidae) and *Oropsylla montana* (Siphonaptera: Ceratophyllidae), Journal of Medical Entomology, tjaa113, https://doi.org/10.1093/jme/tjaa113

LICCIOLI, S., T. STEPHENS, S.C. WILSON, J.M. McPHERSON, L.M. KEATING, K.S. ANTONATION, T.K. BOLLINGER, C.R. CORBETT, D.L. GUMMER, L.R. LINDSAY, T.D. GALLOWAY, T.K. SHURY, and A. MOEHRENSCHLAGER. 2020. Enzootic maintenance of sylvatic plague in Canada's threatened black-tailed prairie dog ecosystem. Bulletin of the Ecological Society of America 101(4):e01741. https://doi.org/10.1002/bes2.1741. 5 pp.

Medvedev, S. G., & Verzhutsky, D. B. (2020). Diversity of Fleas, Vectors of Plague Pathogens: the Flea *Oropsylla silantiewi* (Wagner, 1898) (Siphonaptera, Ceratophyllidae). Entomological Review, 100, 45-57.

Willis, Gabriela, Kerryn Lodo, Alistair McGregor, Faline Howes, Stephanie Williams, Mark Veitch. 2019. New and old hotspots for rickettsial spotted fever acquired in Tasmania, 2012–2017. Aust NZ J Public Health. 2019; 43:389-94; doi: 10.1111/1753-6405.12918.

Objective: To describe the epidemiology and clinical characteristics of Tasmania-acquired rickettsial disease notified to the Department of Health in Tasmania from 2012 to 2017 inclusive.

Methods: Data on rickettsiosis cases acquired and notified in Tasmania between 1 January 2012 and 31 December 2017 were analysed descriptively.

Results: Eighteen cases of rickettsial infection notified in Tasmania 2012–17 and likely acquired in the state met one of three case definitions: 12 confirmed (67%), four probable (22%), and two possible (11%). The mean number of cases per year was 3.0 (population rate 0.6 per 100,000 population/year); 60% of cases occurred in November and December. Cases were more commonly older males. Fever, lethargy, and rash were commonly reported symptoms. Thirteen cases were likely acquired on Flinders Island, three around Great Oyster Bay and two in the Midlands.

Conclusions: This study extends our knowledge of the epidemiology of rickettsial disease in Tasmania. This is the first account including confirmed cases acquired in the Midlands of Tasmania.

Implications for public health: Increased knowledge and awareness of epidemiology of rickettsial infection in Tasmania is essential for timely diagnosis and appropriate treatment. These findings bear wider relevance outside Tasmania because visitors may also be at risk.

Key words: *Rickettsia*, infectious diseases, epidemiology, surveillance, Flinders Island spotted fever.

Microbial Symbiotes

Betancourt-Ruiz, P., Martínez-Díaz, H. C., Gil-Mora, J., Ospina, C., Olaya-M, L. A., Benavides, E., ... & Hidalgo, M. (2020). *Candidatus Rickettsia senegalensis* in Cat Fleas (Siphonaptera: Pulicidae)

Collected from Dogs and Cats in Cauca, Colombia. Journal of medical entomology, 57(2), 382-387.

Bezerra-Santos, M. A., Nogueira, B. C. F., Yamatogi, R. S., Ramos, R. A. N., Galhardo, J. A., & Campos, A. K. (2020). Ticks, fleas and endosymbionts in the ectoparasite fauna of the black-eared opossum *Dipelphis aurita* in Brazil. Experimental and Applied Acarology, 80(3), 329-338.

Buhler, Kayla J., Ricardo G. Maggi, Julie Gailius, Terry D. Galloway, Neil B. Chilton, Ray T. Alisauskas, Gustaf Samelius, Émilie Bouchard and Emily J. Jenkins. 2020. Hopping species and borders: detection of *Bartonella* spp. in avian nest fleas and arctic foxes from Nunavut, Canada. Parasites Vectors (2020) 13:469. https://doi.org/10.1186/s13071-020-04344-3

Background: In a warmer and more globally connected Arctic, vector-borne pathogens of zoonotic importance may be increasing in prevalence in native wildlife. Recently, *Bartonella henselae*, the causative agent of cat scratch fever, was detected in blood collected from arctic foxes (*Vulpes lagopus*) that were captured and released in the large goose colony at Karrak Lake, Nunavut, Canada. This bacterium is generally associated with cats and cat fleas, which are absent from Arctic ecosystems. Arctic foxes in this region feed extensively on migratory geese, their eggs, and their goslings. Thus, we hypothesized that a nest flea, *Ceratophyllus vagabundus vagabundus* (Boheman, 1865), may serve as a vector for transmission of *Bartonella* spp.

Methods: We determined the prevalence of *Bartonella* spp. in (i) nest fleas collected from 5 arctic fox dens and (ii) 37 surrounding goose nests, (iii) fleas collected from 20 geese harvested during arrival at the nesting grounds and (iv) blood clots from 57 adult live-captured arctic foxes. A subsample of fleas were identified morphologically as *C. v. vagabundus*. Remaining fleas were pooled for each nest, den, or host. DNA was extracted from flea pools and blood clots and analyzed with conventional and real-time polymerase chain reactions targeting the 16S-23S rRNA intergenic transcribed spacer region.

Results: *Bartonella henselae* was identified in 43% of pooled flea samples from nests and 40% of pooled flea samples from fox dens. *Bartonella vinsonii berkhoffii* was identified in 30% of pooled flea samples collected from 20 geese. Both *B. vinsonii berkhoffii* (n = 2) and *B. rochalimae* (n = 1) were identified in the blood of foxes.

Conclusions: We confirm that *B. henselae, B. vinsonii berkhoffii* and *B. rochalimae* circulate in the Karrak Lake ecosystem and that nest fleas contain *B. vinsonii* and *B. henselae* DNA, suggesting that this flea may serve as a potential vector for transmission among Arctic wildlife.

Keywords: Arctic fox, Bartonella, Disease ecology, Flea, Geese, Nunavut, Vector-borne disease, Wildlife, Zoonotic

Calvani, N. E., Bell, L., Carney, A., De La Fuente, C., Stragliotto, T., Tunstall, M., & Šlapeta, J. (2020). The molecular identity of fleas (Siphonaptera) carrying *Rickettsia felis, Bartonella clarridgeiae* and

Bartonella rochalimae from dogs and cats in Northern Laos. Heliyon, 6(7), e04385.

Eremeeva, M. E., Capps, D., McBride, C. L., Williams-Newkirk, A. J., Dasch, G. A., Salzer, J. S., ... & Durden, L. A. (2020). Detection of *Rickettsia asembonensis* in Fleas (Siphonaptera: Pulicidae, Ceratophyllidae) Collected in Five Counties in Georgia, United States. Journal of Medical Entomology. Journal of Medical Entomology, Volume 57, Issue 4, July 2020, Pages 1246–1253, https://doi.org/10.1093/jme/tjaa029

We conducted a molecular survey of *Rickettsia* in fleas collected from opossums, road-killed and live-trapped in peridomestic and rural settings, state parks, and from pet cats and dogs in Georgia, United States during 1992–2014. The cat flea, *Ctenocephalides felis* (Bouché) was the predominant species collected from cats and among the archival specimens from opossums found in peridomestic settings. *Polygenis gwyni* (Fox) was more prevalent on opossums and a single cotton rat trapped in sylvatic settings. Trapped animals were infested infrequently with the squirrel flea, *Orchopeas howardi* (Baker) and *C. felis*. TaqMan assays targeting the BioB gene of *Rickettsia felis* and the OmpB gene of *Rickettsia typhi* were used to test 291 flea DNAs for *Rickettsia*. A subset of 53 *C. felis* collected from a cat in 2011 was tested in 18 pools which were all bioB TaqMan positive (34% minimum infection prevalence). Of 238 fleas tested individually, 140 (58.8%, 95% confidence interval [CI]: 52.5–64.9%) DNAs were bioB positive. Detection of bioB was more prevalent in individual *C. felis* (91%) compared to *P. gwyni* (13.4%). Twenty-one (7.2%) were ompB TaqMan positive, including 18 *C. felis* (9.5%) and 3 *P. gwyni* (3.2%). Most of these fleas were also positive with bioB TaqMan; however, sequencing of gltA amplicons detected only DNA of *Rickettsia asembonensis.* Furthermore, only the *R. asembonensis* genotype was identified based on NlaIV restriction analysis of a larger ompB fragment. These findings contribute to understanding the diversity of *Rickettsia* associated with fleas in Georgia and emphasize the need for development of more specific molecular tools for detection and field research on rickettsial pathogens.

Fedele, K., Poh, K. C., Brown, J. E., Jones, A., Durden, L. A., Tiffin, H. S., ... & Machtinger, E. T. (2020). Host distribution and pathogen infection of fleas (Siphonaptera) recovered from small mammals in Pennsylvania. Journal of Vector Ecology, 45(1), 32-44.

Heglasová, I., Víchová, B., & Stanko, M. (2020). Detection of *Rickettsia* spp. in Fleas Collected from Small Mammals in Slovakia, Central Europe. Vector-Borne and Zoonotic Diseases.

Moreno Salas L., M. Espinoza-Carniglia, N. Lizama-Schmeisser, L. G. Torres Fuentes, C. Silva de La Fuente, M. Lareschi and D. González-Acuña. 2020. Molecular detection of *Rickettsia* in fleas from micromammals in Chile. Parasite & Vectores, DOI: 10.21203/rs.3.rs-35524/v1

Melis M., M. Espinoza-Carniglia, E. Savchenko, S. Nava, M. Lareschi. 2020. Molecular detection and identification of *Rickettsia felis* in a *Polygenis* flea (Siphonaptera, Rhopalopsyllidae, Rhopalopsyllinae) associated with cricetid rodents from a rural area in central Argentina. Veterinary Parasitology: Regional Studies and Reports 21 (2020): 100445. https://doi.org/10.1016/j.vprsr.2020.100445

Urdapilleta M., Cicuttin GL, De Salvo MN, Pech May A, Salomon OD & Lareschi, M. 2020. Molecular detection and identification of *Bartonella* in the cat flea *Ctenocephalides felis felis* collected from cats and dogs in a border area in northeastern Argentina. Veterinary Parasitology: Regional Studies and Reports 19: 100361 https://doi.org/10.1016/j.vprsr.2019.100361.

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Book review: 4 books on pandemics

by R.L. Bossard

During the pandemic, I read about pandemics.

Honigsbaum, Mark. 2019. **The pandemic century : one hundred years of panic, hysteria, and hubris.** W. W. Norton & Company, New York.

Khan, Ali. 2016. **The next pandemic : on the front lines against humankind's gravest dangers.** PublicAffairs, New York.

Quammen, David. 2012. **Spillover : animal infections and the next human pandemic.** W.W. Norton & Co., New York.

Shah, Sonia. 2016. **Pandemic : tracking contagions, from cholera to Ebola and beyond.** Sarah Crichton Books, Farrar, Straus and Giroux, New York.

All 4 of these books overlap, covering Ebola, HIV (Human Immunodeficiency Viruses), SARS (Severe Acute Respiratory Syndrome), and West Nile plus numerous others, and are super introductions to the biology of pandemics. Each author has his or her own perspective.

Honigsbaum recounts diverse histories of modern pandemics, such as Spanish Fever (1918), parrot fever (psittacosis, 1929), created by the exotic bird trade, Legionnaires' disease (1976), created by air-conditioning, and Zika (2015), possibly spread by the shipping of mosquitoes. Of additional interest to entomologists is his chapter on plague in Los Angeles (1924). There were limited treatment options for pneumonic plague then, other than sera from immune survivors and plague vaccine was apparently not used. Alexander Fleming would not discover penicillin and other antibiotics until 1928. A puzzle at the time was that the assumed reservoirs, rats, were not found. The actual mammals appear to have been ground squirrels. Honigsbaum discourses on the politics of fear and clampdown.

Khan is a U.S. public health official investigating disease and identifying infectious agents. He discusses outbreaks with which he had first-hand experience. These include food-borne *Salmonella*, anthrax, Sin Nombre Hantavirus, the Hurricane Katrina aftermath, MERS (Middle Eastern Respiratory Syndrome), monkeypox, and a plague infection of a couple in 2002. Public health officials are detectives who work under often-demanding conditions to help manage pandemics.

Quammen focuses on the interspecies' jumps of zoonotic agents. He is a journalist who works extensively in the field with biologists who sample for infectious agents and antibodies. He clarifies the math of pandemics, as well as the field work involved. He describes how molecular studies are elucidating relationships among the 200 *Plasmodium* species affecting reptiles, mammals, and birds; only 4 *Plasmodium* species cause human malaria. He investigates zoonotic viruses in bats that have affected horses (Hendra), humans in mines (Marburg), and pigs (Nipah). He discusses how it is the small mammals, not deer, which are reservoirs for *Borrelia burgdorferi* bacteria and Lyme disease, and the taxonomic controversy between the ticks *Ixodes scapularis* and *I. dammini*, since synonymized.

Shah's book starts off with interspecies jumps, but extends the dialogue by comparing why common factors create pandemics. Even though each pandemic is unique, “history repeats itself”, to an extent. Humans provide propulsion for the microbe, especially through international air travel, immigration, refugees, and medical tourism. That last activity has dispersed the NDM-1 plasmid that creates broad-spectrum antibiotic resistance. She mentions MRSA (Methicillin-Resistant *Staphylococcus aureus*) and Shiga-toxin *E. coli*. Other common factors in pandemics include high population density of humans, lack of cleanliness, politics and corruption, censorship, blame of scapegoats, panic, social unrest, and limitations in cures. She points out that disease outbreaks are affecting other animals, too, in particular, Colony Collapse Disorder (CCD) (insect pollinators), Chytrid Disease (amphibians with the fungi *Batrachochytrium dendrobatidis*), and White Nose Syndrome (bats with *Pseudogymnoascus destructans*). Shah’s special emphasis is cholera, amazingly, a zoonosis from copepod hosts (an aquatic crustacean)!

The authors remark on the lack of a “global health surveillance system”. Surprisingly to me, the World Health Organization (WHO) is not currently such a system. As Shah points out:

> "As of 2013, only 80 of the WHO’s 193 member nations had the surveillance capacity to fulfill their WHO-mandated requirements. Antibiotic-resistant pathogens such as NDM-1 are being found basically by accident. There’s no national surveillance to track the bug in India. In most avian-flu-affected countries, livestock are not monitored for signs of the virus. Nor are humans, for that matter”.

Other measures discussed in these books could help us deal with pandemics. These include establishing natural preserves (see Flea News 83 on setting aside 50% of the Earth's surface for wild ecosystems), and linking humans, livestock, and wild species in a “One Health” (“Ecohealth”) paradigm. Limiting, screening, and taxing at 1% international travel would permit improved disease monitoring and control. Restricting the use of antibiotics, especially in livestock, and as prophylactics for weight gain, would protect our irreplaceable antibiotics. Improving the environment of food animals, and avoiding the international transport and consumption of wild animals for pets, their body parts, medical experimentation, and food, would help bound microbial spillover. Better care in laboratories and other facilities might limit the little-known, but startlingly frequent, containment failures the books narrate.

Fascinating are these books' reportings on the animals and microbes involved and their habitats, the humans who are infected, the fantastic technology, and the keen observations made by dedicated, often puzzled, medical specialists in the field and laboratory. We studied lists in textbooks of infectious diseases with exotic names, but these 4 books really enliven the absorbing stories behind the outbreaks.

Writing before the current Covid-19 pandemic, the authors are wary about guessing on the "next pandemic". The influenzas, like H1N1, are a continuing concern. Fungal infections and arthropod bites are increasing with global warming. Quammen quotes the epidemiologist D.S. Burke, who warned in 1997 that coronaviruses "should be considered as serious threats to human health…" That group includes SARS (2002), MERS (2012), and the ongoing Covid-19.

The books concur that microbes, largely zoonotic, both unknown and long-thought vanquished, have materialized. Pandemics are features of modern life.

R.L. Bossard

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Directory of Siphonapterists (updated)

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Molecular biology of fleas

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Siphonaptera Literature (2020)
See next Flea News.

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Fleas in Art and History

BYPATHS IN DIXIE: Folk Tales of the South
by SARAH JOHNSON COCKE
(b. February 7, 1865, Selma, Alabama, d. January 20, 1944, Roanoke, Virginia)

E·P·Dutton & Company, New York, 1911.

IV. MISS RACE HOSS AN' DE FLEAS (excerpt)

"Po' Sis' Cow beller' an' beller' fur Mister Cow ter come an' run Miss Race Hoss off, but law, Mister Cow bizzy tendin' ter 'es bizness an' he don't hear ole Sis' Cow. Jes' den, Sis' Dog an' Sis' Sow an' Sis' Cat sorter whisper 'mongst deysefs. Pres'ntly dey all jumps up an' starts ter shakin' deyse'fs whensomever Miss Race Hoss git clost ter 'em. Fus' thing yer knows, Miss Race Hoss stop' her fancy steppin' an' holler, 'How 'pon earth come dese fleas ter git on top er me?' She jump' an' she roll', she jump' an' she roll', an' I speck she'd bin er jumpin' an' er rollin' plum tell now, ef dem fleas teeth had er bin strong nuf ter er bit thu Miss Race Hosses hide, but yer see wid all de bitin' dey bin doin', dar wasn't one uv 'em dat got er good clinch on Miss Race Hoss. So Sis' Sow's fleas say dey gwine back home ter vit'als dey wus rais'd on, an' Sis' Dog's fleas say dey wus gwine back whar de meat wus tender, an' Sis' Cat's fleas say dey don't see no use tryin' ter git er livin' off'n hoss hide when dar wus plenty er kitten meat dat would melt in yo' mouf. So wid dat, all uv de fleas give er jump, an' lands back on Sis' Sow an' Sis' Dog an' Sis' Cat; an', honey, dem fleas ain't no sooner jumpt, dan Miss Race Hoss jump, too. She give er back-runnin' start an' wus ov'r dat fence 'fo' you know'd it; an' bless yo' heart, she come mouty nigh ter jumpin' on her own little colt dat had done foller' her onbeknownst. De colt nev'r seed es ma mirate an' car'y on so b'fo', an' he got so occipi'd watchin' her dat he plum fergit ter mention he was dar. Howsomev'r, when Miss Race Hoss come er flyin' ov'r dat fence she come so close ter de little colt dat whil'st he was er gittin' outen de way, he trip' es own sef an' fell er sprawlin' flat.

"Po' little colt commenc' ter whinnyin' an' cryin', an' his ma was so sorry an' miserbul dat she tuck him in her arms an' 'gun ter pattin' an' er singin' ter him jes' like dis:

"'Mama luvs de baby,
Papa luvs de baby,
Ev'ybody luvs de baby,
Hush yo' bye, doan you cry,
Go ter sleepy lill'e baby... "

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(On the next page)

The Head of the Ghost of a Flea, circa 1819

Artist: William Blake
(b. 28 November 1757, Soho, London, England, d. 12 August 1827, Charing Cross, London).
Graphite on paper, 189 x 153 mm.

(Editor's note: a study of *Ghost of a Flea*, Flea News 74 June 2014)

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Commentary on Allen Haydon Benton (1921-2014)

by Terry Galloway

When scientists live many years beyond retirement and discontinue or reduce their research and the endeavours that were central to their careers as scientists, there is always the danger they may lose contact with former colleagues. If they no longer publish in their field and pursue other avenues of interest, the old lines of communication wear thin, especially if they aren't connected via e-mail or social media. Unfortunately, this is what happened in my communication with Allen Benton, life-long naturalist, writer, and flea researcher.

My own interests in fleas arose from the project of a graduate student in my lab on the seasonal host interactions of the American dog tick (*Dermacentor variabilis*) in southern Manitoba. As is so often the case, the first field season for the student in 1979 was a complete disaster as far as ticks were concerned. This was the year of no summer. Small mammal populations were down and tick numbers insignificant. In desperation, the student collected fleas from the mammals and when everything turned around the next season, I inherited the orphaned fleas. When my interests began to include fleas infesting birds, I first encountered the publications of Allen Benton. We struck up correspondence in the early 1980s and continued until, unbeknownst to me, two years before Allen died. When he failed to respond to my annual Christmas/New Year's letters, I assumed he was no longer able to carry on corresponding. I had not heard that he had died, and time gradually passed. I recently wondered what had become of Allen, so I searched the internet and sadly discovered his obituary in *Chautauqua Today*. Allen had died on 29 September, 2014 after a brief illness, at the age of 93. It turned out the obituary originally appeared on 3 October, 2014 in the *Observer*, a newspaper in Dunkirk, New York, where Allen had been a regular contributor for many years. Most of what follows is taken from this obituary and from my own correspondence over the years with Allen.

Allen Benton was born 4 September, 1921 to Haydon W. Benton and Pearl Diddy Benton in Cato, New York. He graduated from Cato High School, then earned BS, MS and PhD degrees from Cornell University. A combat soldier in World War II, he served with the 112th Cavalry Regimental Combat Team when he spent time in New Guinea and saw action in Luzon, the Philippines and Japan. When he was shipped to New Guinea in World War II (1944), he lamented his first three days at sea, during which he spent the entire time in his bunk, deathly seasick. On the fourth day, an officer ordered him on deck for submarine drill, and he was never seasick again, even during heavy seas on the trip home from Yokohama to Seattle.

He taught a variety of courses across many disciplines in biology during his 13 years at State University of New York (SUNY) Albany and 22 years at SUNY Fredonia (three years as chair of the Biology Department), including Introductory Entomology, Insect Ecology, Insect Taxonomy, and Conservation Biology, among others. He was a co-author with William E. Werner, Jr. of the highly successful, <u>Manual of Field Biology and Ecology</u>, published by McGraw-Hill. At Fredonia, he was the inaugural recipient of the New York State Distinguished Teacher Award, and he certainly influenced the interests and careers of many of his students. Diane R. Chodan, former student and Lifestyles editor of the *Observer*, published a heart-warming account of Allen in the paper on 3 November, 2014. He once wrote to me, "Isn't it a joy to have a class of enthusiastic students! My approach to teaching was geared to open-ended learning to motivated students, and once or twice in my 35 years of teaching I got whole classes who were eager to learn". Dr. Shari Yudenfreund-Sujka graduated *magna cum laude* from SUNY Fredonia in 1978 with a degree in Biology, and earned her Ph.D. from St. George's University School of Medicine. She credited Allen Benton and Kevin Fox for stimulating her to make a career in science, and dedicated the head's office in the new SUNY Science Center to their honor on 22 January, 2015.

Allen was an avid reader who loved poetry, especially Haiku, and wrote a number of volumes of poems, sometimes under his own name (The Wheel of Life, Haiku by Followers of Basho, 2003) and sometimes under a pseudonym, Albert Ezra Fitzwarren (Sonnets from Nebraska and Beyond, 1984; Slivers of Jade, 1986; The Nature of Nature from Alpha to Zeta, 1991, each with an introduction by Casper C. Abernathy, aka Allen H. Benton). His love of poetry and the written word no doubt contributed to his ability to lengthy passages and his fondness for puns. He published at least two novels (Of Time Ill Spent. Being the Early Life and Times of Jamison B. Carruthers, and Bronski's Medals. A Novel of World War II). Allen wrote nature guides (Key to Common Ferns: Identifications of 28 Species of Common and Widespread Northeastern Ferns, Key to the Mammals of Western New York, Key to the Fleas of the Eastern United States, Fleas of Medical and Veterinary Importance), and at least one series of children's booklets with Richard L. Bunting (Young People's Nature Guides: Eleven 16-page booklets to introduce 7-10-year-olds to outdoor activities and the beauty of nature). He appeared on television in a weekly show on bird-watching on WRGB TV in Albany and published a weekly newspaper column, "On the Wing," in the Knickerbocker News. In Fredonia, his weekly nature column, "World of the Wild," appeared for two decades in the Dunkirk *Evening Observer*. His obvious passion and love for the natural world and for conveying this passion to the public led to publication of Wild Worlds in 1988, a compilation of his articles in the *Observer*. More than 200 columns later, and since Wild Worlds was so well received, he published a sequel, Light and Natural in 1992. Birding Through Life: Wanderings of A Born Birder (2004) and To Walk in Beauty: The Way of the Naturalist (2005) were published later in life, and are stories told by Allen about his experiences as a naturalist in the field.

For readers of *Flea News*, Allen's career as a siphonapterist is of greatest interest. I'm afraid the question of how he became interested in fleas is one I neglected to ask in our correspondence. Perhaps he encountered fleas during his Ph.D. studies on small mammals at Cornell University ("A study of the northern pine mouse, *Pitymys pinetorum scalopsoides* (Audubon and Bachman)". 1953. 66 pp.). However, in his 1955 publication on the biology of the pine mouse (*Journal of Mammalogy* 36, p. 60), he described finding lots of the sucking louse, *Hoplopleura acanthopus* (Burmeister), but not one flea on more than one hundred mice. He certainly showed an interest in parasitology based on this, and his 1954 paper was on *Moniliformis clarki* (Ward)[1] in eastern New York (*Journal of Parasitology* 40: 102-103). In any case, his first refereed papers on fleas appeared in 1955, when he expanded on the descriptions and taxonomic status of *Epitedia w. wenmanni* (Rothschild, 1904) and based on comments by Hopkins, what he referred to as *Epitedia w. testor* (Rothschild, 1915) (Benton 1955). For this study, he examined 544 specimens from across North America, and in the process no doubt launched a career-long correspondence with many colleagues acknowledged in the paper, including G.H.E. Hopkins, F.G.A.M. Smit, Harold Stark, E.W. Jameson, Jr., William Jellison, Richard Eads, and George Holland, among others. In this paper, too, there was a hint of Allen's future interest in the seasonal patterns and life history of fleas. In the same year, he published a paper on a method for preparing fleas onto slides and one on rarely collected fleas in New York State.

Allen donated his collection of 4000+ slides of mammal fleas and much of his reprint collection to the Florida State Collection of Arthropods in Gainesville. The majority of his bird fleas (1348 specimens) was sent to me in 1993 and was deposited in the J.B. Wallis/R.E. Roughley Museum of Entomology in the Department of Entomology at the University of Manitoba. Allen shipped the slides to me in several parcels, the first of which I hope was treated with good humour among Canada Customs authorities. Inside the parcel, Allen included a short note to me in which he said, “Sometimes I have trouble with shipments of fleas which had to cross international borders, so I am going to send them a few at a time. I will label them “Microscope Slides” and hope that no misguided customs official will feel that they shouldn’t be admitted to Canada.” Well, on the customs declaration form, he indicated the value of the shipment as $100, an amount that caused Canada Customs officials to open the parcel and find Allen’s note to me. When the parcel arrived in Winnipeg, I found numerous photocopies of his note, on one of which the pertinent paragraph was circled in bold, red highlighter. The parcel took an inordinate length of time to arrive on my desk, and I was charged $8.72 Goods and Services Tax plus a $5.00 Handling Fee. Allen and I, and I’m sure several Customs officers, all had a good chuckle.

I think Allen’s is a story of a life well lived. One of my lasting regrets is that I was never able to make the journey to Fredonia to meet him in person.

Acknowledgements

Special thanks to Dave Rowley, News Director at WDOE/KIX Country in Dunkirk, NY, and Gregory Bacon, Managing Editor of the *Observer* in Dunkirk for permission to extract vital information about Allen from his obituary. Thanks also to Drs Paul Skelley and Gary Steck, Systemic Entomologists at the Florida State Collection of Arthropods, Gainesville, Florida for their assistance in tracking down Allen’s donation of mammal fleas. Carol Galloway cast her editorial eye over the final draft of this obituary.

Editor's Footnote. 1) *Moniliformis clarki* (Ward) is an intestinal endoparasite in the acanthocephalan phylum (spiny-headed worm). Abbreviated obituary of Allen H. Benton appears in Flea News 85 March 2020. List of his publications on Siphonaptera appears in Flea News 83 December 2018.